Ho Young Jeong, Ph.D.

Assistant Professor,
Kyung Hee University
ghy27@khu.ac.kr

Employment

Assistant Professor	Kyung Hee University, Mar 2026 – present
Postdoctoral Researcher	Korea University, Nov 2023 – Mar 2026
Postdoctoral Researcher	Sogang University, April 2022 – Oct 2023

Education

Ph.D. Purdue University, West Lafayette, IN, United States, May 2022
Industrial Management Engineering

M.S. Purdue University, West Lafayette, IN, United States, May 2018
Industrial Management Engineering

B.S. Inha University, Incheon, Republic of Korea, July 2016
Bachelor of Industrial Management Engineering

3 Major Achievements

[1] **Jeong, H. Y.**, Song, B. D. 2025. Optimizing Urban Logistics: VRP with Underground Transportation. IEEE Transactions on Intelligent Transportation Systems.

[2] **Jeong, H. Y.**, and Song, B. D., 2025. Meta-Learning-Based Adaptive Operator Selection for Traveling Salesman Problem. Applied Soft Computing.

[3] **Jeong, H. Y.**, and Song, B. D., 2025. Optimization of Urban Logistics with Multi-Modal Systems: A Comprehensive Study of the Airship-Vehicle Routing Problem. Transportation Research Part E: Logistics and Transportation Review.

Publications

Journals

J1. **Jeong, H. Y.**, and Song, B. D., 2025. Optimization of Urban Logistics with Multi-Modal Systems: A Comprehensive Study of the Airship-Vehicle Routing Problem. Transportation Research Part E: Logistics and Transportation Review, 204, 104415. (SCIE, JCR 1.4%)

J2. **Jeong, H. Y.**, and Song, B. D., 2025. Meta-Learning-Based Adaptive Operator Selection for Traveling Salesman Problem. Applied Soft Computing, 113930.

J3. **Jeong, H. Y.**, and Song, B. D., 2025. Optimizing Urban Logistics: Vehicle Routing Problem with Underground Transportation. IEEE Transactions on Intelligent Transportation Systems. (SCIE, JCR 2.5%)

J4. Kim, Y., **Jeong, H. Y.***, and Lee, S., 2024. Drone delivery problem with multi-flight level: Machine learning based solution approach. Computers & Industrial Engineering, 197, p.110565. (SCIE)

J5. Guo, F., **Jeong, H. Y.**, Park, D., Kim, G., Sung, B., and Kim, N., 2024. Numerical optimization of variable blank holder force trajectories in stamping process for multi-defect reduction. Materials, 17(11), p.2578. (SCIE)

J6. Guo, F.[1], **Jeong, H. Y.**[1], Park, D., Sung, B., and Kim, N*., 2024. Numerical multi-objective optimization of segmented and variable blank holder force trajectories in deep drawing based on DNN-GA-MCS strategy. The International Journal of Advanced Manufacturing Technology, pp.1-24. (SCIE)

J7. Kim, Y., **Jeong, H. Y.**, J. Park, K. Kim, H. Kwon, G. Ju, and N. Kim, 2023. Optimizing process parameters for hot forging of Ti-6242 alloy: A machine learning and FEM simulation approach. Journal of Materials Research and Technology, 27, pp. 8228-8243. (SCIE, JCR 6.1%)

J8. Kim, B., **Jeong, H. Y.***, and Lee, S., 2023. Two-echelon collaborative routing problem with heterogeneous crowd-shippers. Computers & Operations Research, 160, p.106389. (SCIE)

J9. **Jeong, H. Y.** and Choi, C.*, 2023. Adaptive supply chain system design for fruit crops under climate change. Systems, 11(10), p.514. (SSCI)

J10. **Jeong, H. Y.** and Lee, S.*, 2023. Drone routing problem with truck: Optimization and quantitative analysis. Expert Systems with Applications, 227, p.120260. (SCIE, JCR 5.2%)

J11. Park, J., Kim, Y., **Jeong, H. Y.**, Kwon, H., Kwon, Y., and Kim, N.*, 2023. Cogging process design of M50 bearing steel for billet quality. Journal of Materials Research and Technology, 26, pp.5576-5593. (SCIE, JCR 6.1%)

J12. **Jeong, H. Y.***, Kim, Y., Lee, S., Moreland, J., and Zhou, C., 2023. Disruption propagation and repair response in interdependent systems: Network model and simulation approach. Simulation Modelling Practice and Theory, 124, p.102730. (SCIE)

J13. **Jeong, H. Y.**, Park, J., Kim, Y., Shin, S. Y., and Kim, N.*, 2023. Processing parameters optimization in hot forging of AISI 4340 steel using instability map and reinforcement learning. Journal of Materials Research and Technology, 23, pp.1995-2009. (SCIE, JCR 6.1%)

J14. **Jeong, H. Y.**, Song, B. D.*, and Lee, S., 2022. Optimal scheduling and quantitative analysis for multi-flying warehouse scheduling problem: Amazon airborne fulfillment center. Transportation Research Part C: Emerging Technologies, 143, p.103831. (SCIE, JCR 9.0%)

J15. **Jeong, H. Y.**, Song, B. D.*, and Lee, S., 2020. The flying warehouse delivery system: A quantitative approach for the optimal operation policy of airborne fulfillment center. IEEE Transactions on Intelligent Transportation Systems, 22(12), pp.7521-7530. (SCIE, JCR 2.5%)

J16. **Jeong, H. Y.**, David, J. Y., Min, B. C., and Lee, S.*, 2020. The humanitarian flying warehouse. Transportation Research Part E: Logistics and Transportation Review, 136, p.101901. (SCIE, JCR 1.4%)

J17. **Jeong, H. Y.**, Song, B. D.*, and Lee, S., 2019. Truck-drone hybrid delivery routing: Payload-energy dependency and No-Fly zones. International Journal of Production Economics, 214, pp.220-233. (SCIE, JCR 2.4%)

Working papers

- (Under review) Choi, C., and **Jeong, H. Y.***, under review. Integrated approach for long term capacity expansion and operational scheduling of grid system: case study in MISO Region. Sustainable Energy, Grids and Networks

- (In preparation) **Jeong, H. Y.**, and Song, B. D., Tariff-Aware Heuristic Optimization for Integrated Vehicle Routing and V2G/G2V Scheduling in Grid-Connected Electric Fleets. IEEE Transactions on Transportation Electrification.

Conference Proceedings

C1. Kim, B., Jeong, H.Y. and Lee, S., Last-Mile Delivery Capacity Building in Pandemics Utilizing Community Resources. Available at SSRN 4168513.

C2. Kim, Y., Jung, H. and Lee, S., 2021, August. Drone Delivery Vehicle Routing Problem with Multi-flight Level. In IFIP International Conference on Advances in Production Management Systems. Cham: Springer International Publishing. pp. 43-51

C3. Jeong, H.Y. and Lee, S., 2021. Collaborative Hybrid Delivery System: Drone Routing Problem Assisted by Truck. In Advances in Production Management Systems. Artificial Intelligence for Sustainable and Resilient Production Systems: IFIP WG 5.7 International Conference, APMS 2021, Nantes, France, September 5–9, 2021, Proceedings, Part III. Springer International Publishing. pp. 33-42

C4. Kim, Y., Jeong, H.Y., Park, J., Kim, K., Kwon, H., Ju, G. and Kim, N., Optimizing Process Parameters for Hot Closed Forging of Ti-6242 Alloy: A Machine Learning and Fem Simulation Approach. Available at SSRN 4588071.

C5. Song, B.D., Jun, S., Jung, H.Y. and Lee, S., 2019. Movable Unmanned Aerial System: Optimization of System, Resource Design and Drone Routing. Procedia Manufacturing, 39, pp.300-306.

C6. Jeong, H.Y. and Lee, S., 2019. Optimization of Vehicle-Carrier Routing: Mathematical Model and Comparison with Related Routing Models. Procedia Manufacturing, 39, pp.307-313.

Presentation

P1. **Jeong, H. Y.**, and Lee, S., 2019. Airship-based drone delivery system: Quantitative approach for managerial and operational guidelines. Institute for Operations Research and the Management Sciences (INFORMS) Annual Meeting, Seattle, US.

P2. **Jeong, H. Y.**, and Lee, S., 2019. Optimization of Vehicle-Carrier Routing Mathematical Model and Comparison with Related Routing Models. International Conference on Production Research 25th (ICPR25), Chicago, US.

P3. Song, B. D., Jun, S., **Jeong, H. Y.**, and Lee, S., 2019. Movable Unmanned Aerial System Optimization of System Resource Design and Drone Routing. International Conference on Production Research 25th (ICPR25), Chicago, US.

P4. **Jeong, H. Y.**, and Lee, S., 2018. Vehicle-Carrier Routing Problem. Institute for Operations Research and the Management Sciences (INFORMS) Annual Meeting, Phoenix, US.

P5. **Jeong, H. Y.**, and Lee, S., 2018. Scheduling Hybrid Delivery System of Truck and Drone: Energy-Payload Dependency and No-Fly Zone. Institute of Industrial and Systems Engineers (IISE) Annual Meeting, Orlando, US.

Research Experience

R1. (PI) Optimization of Smart V2G-Based Integrated Logistics and Power Grid Operations, Jan 2025 –
Funded by the Ministry of Education (NRF-02311960)

R2. Real-time Last-Mile Delivery Using Collaborative Ground-Aerial Movable Stations, Jan 2023 –
Funded by the Ministry of Science and ICT

R3. Development of Multipoint Servocontrol Die Cushion and Process Monitoring Module, Dec 2023
Funded by the Ministry of Trade, Industry and Energy; Associated with (주)한빛산업

R4. Motor Housing for High-Quality Electric Vehicles through High-Function Molds, Dec 2023
Funded by the Ministry of SMEs and Startups; Associated with (주)대국정공

R5. Development of Material Forging and Manufacturing Process Technology, Dec 2023
Funded by the Ministry of Trade, Industry and Energy; Associated with (주)케이피에프충주공장

R6. Development of Vibration Insulation Products and Process Technology for PBV, Jun 2023
Funded by the Ministry of Trade, Industry and Energy; Associated with (주)코다코

R7. Development of Multipoint Servocontrol Die Cushion and Process Monitoring Module, Dec 2022
Funded by the Ministry of Trade, Industry and Energy; Associated with (주)한빛산업

R8. Development of Manufacturing Technology for Multilateral Body Parts, Dec 2022
Funded by the Ministry of Trade, Industry and Energy; Associated with (주)새한산업

R9. Development of Material Forging and Manufacturing Process Technology, Dec 2022
Funded by the Ministry of Trade, Industry and Energy; Associated with (주)케이피에프충주공장

R10. Development of Integrated Housing for Powertrain with High Cooling Performance, Dec 2022
Funded by the Ministry of Trade, Industry and Energy; Associated with (주)코넥

R11. Development of Vibration Insulation Products and Process Technology for PBV, Dec 2022
Funded by the Ministry of Trade, Industry and Energy; Associated with (주)코다코

R12. Visualization of Repair Operations Management for Networked Systems Resilience, May 2022
Funded by Navy Crane Center, IN, United States

R13. Resilience in Networked Systems Using Collaboration (RNSC) Project, May 2019
Funded by Navy Crane Center, IN, United States

Skills

Python, R, MATLAB, C#, Gurobi, CPLEX, PyTorch, TensorFlow, Abaqus, DEFORM, ARENA, Unity

Academic Award

The 1st Runner-up paper on International Conference on Production Research 25th (ICPR25), 2019